# Influenza A H7N9, China, 2013

## Overview

This dataset contains individual-level data on 136 confirmed human cases of avian influenza A (H7N9) in China, 2013. It includes dates of onset, hospitalisation, and outcome, along with patient demographics (age, gender) and province. The H7N9 subtype emerged in early 2013 with zoonotic spillover from poultry, causing severe respiratory illness with a high case fatality rate.

Source: Kucharski AJ, Mills HL, Donnelly CA, Riley S (2014). Transmission potential of healthcare-acquired infections in acute care settings. *PLOS Currents Outbreaks*. Data archived at Dryad: doi:10.5061/dryad.2g43n.

## Loading the data

```julia
using DataFrames
using Outbreaks
df = Outbreaks.fluH7N9_china_2013()
size(df)
```

```
(136, 8)
```

## First few rows

```julia
first(df, 6)
```

| | case_id | date_of_onset | date_of_hospitalisation | date_of_outcome | outcome | gender | |
|---|---|---|---|---|---|---|---|
| | Int64 | String15 | String15 | String15 | String7 | String3 | |
| 1 | 1 | 2013-02-19 | NA | 2013-03-04 | Death | m | ... |
| 2 | 2 | 2013-02-27 | 2013-03-03 | 2013-03-10 | Death | m | ... |
| 3 | 3 | 2013-03-09 | 2013-03-19 | 2013-04-09 | Death | f | ... |
| 4 | 4 | 2013-03-19 | 2013-03-27 | NA | NA | f | ... |
| 5 | 5 | 2013-03-19 | 2013-03-30 | 2013-05-15 | Recover | f | ... |
| 6 | 6 | 2013-03-21 | 2013-03-28 | 2013-04-26 | Death | f | ... |

## Summary statistics

```julia
describe(df)
```

| | variable | mean | min | median | max | nmissing | eltype |
|---|---|---|---|---|---|---|---|
| | Symbol | Union... | Any | Union... | Any | Int64 | DataType |
| 1 | case_id | 68.5 | 1 | 68.5 | 136 | 0 | Int64 |
| 2 | date_of_onset | | 2013-02-19 | | NA | 0 | String15 |
| 3 | date_of_hospitalisation | | 2013-03-03 | | NA | 0 | String15 |
| 4 | date_of_outcome | | 2013-03-04 | | NA | 0 | String15 |
| 5 | outcome | | Death | | Recover | 0 | String7 |
| 6 | gender | | NA | | m | 0 | String3 |
| 7 | age | | 15 | | ? | 0 | String3 |
| 8 | province | | Anhui | | Zhejiang | 0 | String15 |

## Outcomes

```
using StatsBase
outcome_counts = combine(groupby(df, :outcome), nrow => :count)
for row in eachrow(outcome_counts)
    println("$(rpad(coalesce(string(row.outcome), "missing"), 15)) $(row.count)")
end
```

```
Death          32
NA             57
Recover        47
```

## Epidemic curve

```
onset = dropmissing(df, :date_of_onset)
sort!(onset, :date_of_onset)
println("Onset range: $(minimum(onset.date_of_onset)) to $(maximum(onset.date_of_onset))")
daily = combine(groupby(onset, :date_of_onset), nrow => :cases)
sort!(daily, :date_of_onset)
for row in eachrow(daily)
    bar = repeat("█", row.cases)
    println("$(row.date_of_onset) | $bar $(row.cases)")
end
```

```
Onset range: 2013-02-19 to NA
2013-02-19 | █ 1
2013-02-27 | █ 1
2013-03-07 | █ 1
2013-03-08 | █ 1
2013-03-09 | █ 1
2013-03-13 | █ 1
2013-03-17 | █ 1
2013-03-19 | ██ 2
2013-03-20 | ██ 2
2013-03-21 | ██ 2
2013-03-22 | █ 1
2013-03-25 | ██ 2
2013-03-27 | █ 1
2013-03-28 | ████ 4
2013-03-29 | ██████ 6
2013-03-30 | ██ 2
2013-03-31 | ███ 3
2013-04-01 | ██████ 6
2013-04-02 | ██ 2
2013-04-03 | ███████ 7
2013-04-04 | ███ 3
2013-04-05 | ██ 2
2013-04-06 | ███████ 7
2013-04-07 | █ 1
2013-04-08 | ██████ 6
2013-04-09 | ███ 3
2013-04-10 | ██████ 6
2013-04-11 | ██████ 6
2013-04-12 | ██████ 6
2013-04-13 | █████ 5
2013-04-14 | █████ 5
2013-04-15 | ████ 4
2013-04-16 | ███ 3
2013-04-17 | █████ 5
```

```
2013-04-18 | ███ 3
2013-04-19 | ██ 2
2013-04-21 | ██ 2
2013-04-23 | ██ 2
2013-04-25 | █ 1
2013-04-26 | █ 1
2013-04-27 | █ 1
2013-04-29 | █ 1
2013-05-03 | █ 1
2013-05-21 | █ 1
2013-07-10 | █ 1
2013-07-27 | █ 1
NA | ██████████ 10
```